

FITNESS

How to Use Heart Rate Variability Data in Your Training

Heart rate variability shows you how well your body is recovered, if you're overtraining, and if you need to improve how you handle stress.

Advice to improve your movement, fitness, and overall health from the #1 in orthopedics in the U.S.

You're probably familiar with heart rate monitoring during a workout. But there's a separate marker of the heart's capacity to endure stress—a measure known as heart rate variability—that might be an even better marker of fitness readiness and fatigue levels.



"Heart rate variability is a measure of stress and recovery within the body, from a physiological standpoint," says [Vincent Luppino, PT, DPT, OCS](#), a physical therapist at HSS Paramus.

This biomarker, when measured accurately with the help of a heart rate variability tracker, can be used on a daily basis to know if your body is primed for a hard workout or if you need to lower the intensity—or rest.

What Is Heart Rate Variability?

Your heart rate is a measure of how many times your heart beats within one minute. This can be counted by checking your pulse on your wrist or neck and counting for one minute or by using a heart rate monitor like a fitness watch. An average heart rate for a normal, healthy person is between 60 to 100 beats per minute (BPM) at rest; for an endurance athlete, that number might even drop to 30 to 40 BPM.

Heart rate variability, or HRV, is the measure of the variation in time between heartbeats, measured in milliseconds. For example, sometimes your heart might beat every 1.2 seconds; other times, it might beat at 0.8 seconds. You need a device to measure that variance in timing (more on that below). Ideally, you'd monitor your HRV for two to five minutes to see what your average is.

A longer HRV, with more time between heartbeats, is in general more desirable than a shorter HRV because it shows your body can recover faster from stress and that the heart can resume its normal rate after having been sped up due to exercise, stress, illness or something else. It's an indicator of how your body handles stress, which includes your training efforts.

How to Measure Heart Rate Variability

If you're a fan of certain exercise classes, you'll often hear instructors direct you to a target heart rate. During exercise, a BPM of 120 all the way to 150 is normal during cardio or endurance training.

The gold standard for measuring heart rate variability is an EKG done by a medical professional. After that, the HRV measures within many fitness trackers are thought to be highly reliable at tracking it. Here are some general tips for measurement:

- The best time to get your HRV is in the morning when you first wake up because you'll be rested (hopefully!) and relaxed, suggests Luppino. A normal HRV at rest for someone in their 20s is 55 to 105; for someone in their 60s, it's 25 to 45.
- Some HRV monitors come with a chest strap, while others can be worn on your wrist or finger in the form of a ring. Not all heart rate monitors have a heart rate variability tracking component, so check your device or do your research if you're looking to buy a fitness tracker that measures it.
- Measuring for at 5 least minutes, if done in a medical setting, should give you a good average, or over the course of a 24-hour period with a heart rate variability monitor you wear at home. "The more information, the better because it encapsulates the entire experience of the heart within a day's timeframe," says Luppino.
- If you're using a heart rate variability tracker for the first time, look at your HRV average over the course of a week to learn your baseline. Once you learn your baseline, you'll have a better idea of what a high or low HRV is for you, then can take steps to modify your routine depending on how you're tracking. "It is better to follow your own individual trend of HRV over time to help you make decisions on any given day," he adds.

Factors Impacting Heart Rate Variability

HRV is impacted by a few things, including:

- **Gender:** We know gender influences HRV but reports are controversial. Men tend to show higher HRV numbers than women, but some studies have shown the opposite to be true.
- **Age:** Your HRV decreases as you age. A 25-year-old male's normal heart rate variability might be 50 to 100 milliseconds. As that person nears middle age, the number could drop to a range of 35 milliseconds to 60 milliseconds.
- **Hormones:** A woman who is tracking her HRV might notice changes at various times throughout the month when she's menstruating.
- **Stress:** When you experience stress, the heart has to pump faster. That means there's less time in between beats, resulting in a shorter HRV.
- **Sleep:** Your body's 24-hour clock plays a role in HRV. You'll find the average number changes throughout the day, as well as if you are tired.
- **Core body temperature:** If your body temperature changes when you aren't feeling well, this can impact your HRV.
- **Metabolism:** Someone who has a metabolic disorder will likely have a lower HRV. Alternatively, the healthier and fitter you are, the likelier you'll have a higher HRV.

How to Use Heart Rate Variability to Plan Your Workouts

It's helpful to know your HRV, but only if you know what to do with it. "Take stock of your heart rate variability day to day and notice patterns over time," suggests Luppino. If you notice a high HRV day, that's the time to go harder with training. "When I see a lower HRV number on my monitor, I'll opt for stretching and breathwork to reduce stress and go to bed early that night," he adds.

Is your HRV a number you should concern yourself with during your workout? Luppino says no. "When I'm working out, I'm not looking at heart rate variability," he says. "I'm looking at my heart rate to modify what zone I'm in so I can focus on hitting my max heart rate during the training session."

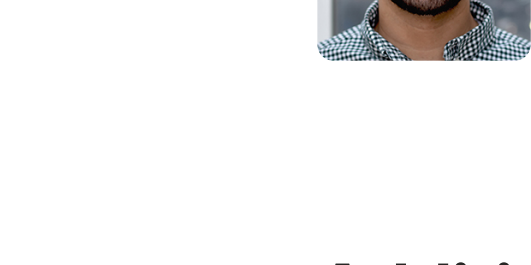
To improve your HRV, practice managing stress with breathing techniques. Your sympathetic nervous system becomes more active when you're feeling stressed, causing the heartbeat to quicken and resulting in a lower HRV. Multiple days of recovery and less stress on the body over time will improve your baseline HRV.

To help release some of that stress, Luppino recommends diaphragmatic breathing, which involves focusing on breathing into the diaphragm (a muscle in your belly) to slow the heartbeat. Try this technique, called box breathing:

- Breathe into your belly for 4 seconds.
- Hold for 4 seconds.
- Breathe out for 4 seconds.
- Hold for 4 seconds.

Published 8/22/2023

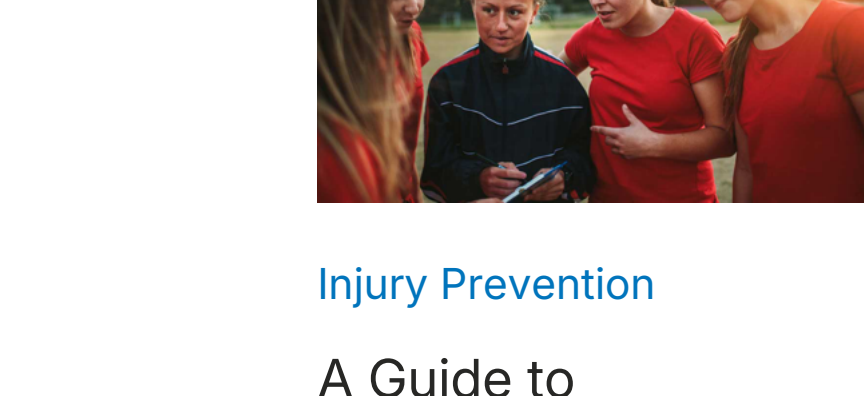
About the Expert



Vincent J. Luppino, PT, DPT, OCS, Cert. DN, COMT, CSCS

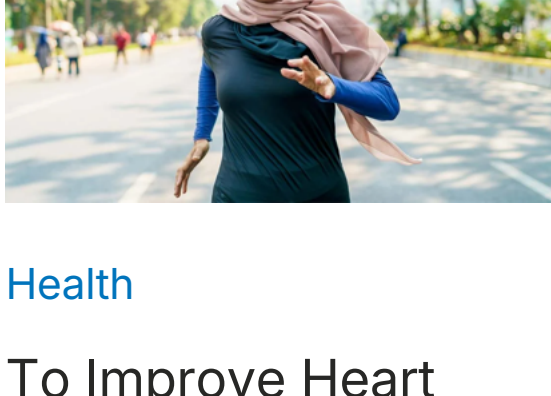
Vincent Luppino is a physical therapist with HSS Rehabilitation

Additional Resources



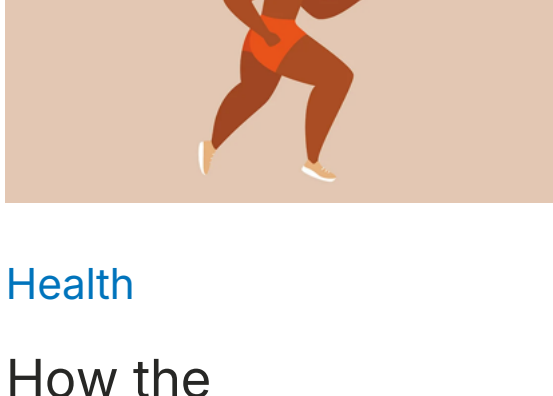
Injury Prevention

A Guide to Neuromuscular Training



Health

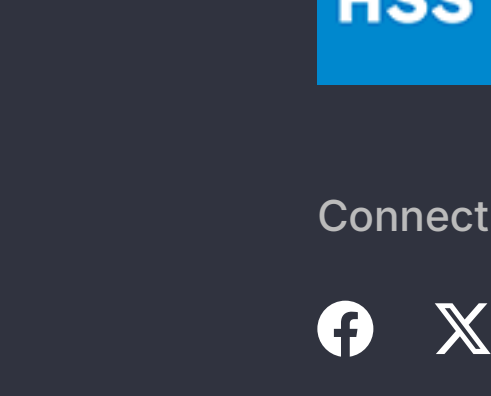
To Improve Heart Health, Start Slowly, Listen to Your Body



Health

How the Parasympathetic Nervous System Can Lower Stress

Make an Appointment at HSS



Connect With Us



Share Your Story

Read Articles from Move Better Feel Better

Subscribe to our eNewsletters

Make a Donation

Learn More

About HSS

Newsroom

Community Benefit

HSS Store

Contact Us

Information For

Patients and Visitors

Medical Professionals

Job Seekers

Employers

Vendors

HSS Staff

Policies

HSS Patient Policies and Guidelines

Website Terms of Use

Website/App Privacy Policy

© 2026 Hospital for Special Surgery, 535 East 70th Street, New York, NY 10021

Hospital for Special Surgery, HSS and the HSS logo are trademarks or registered trademarks of Hospital for Special Surgery in the United States and other countries.

